A Systems-Level Trade Study of CubeSat Deorbit Strategies Under Mass and Power Constraints: Propulsion, Atmospheric Drag and Solar Activity Effects

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1. Introduction

The rapid growth of small satellite missions has significantly increased congestion in low Earth orbit (LEO), raising concerns about long-term orbital sustainability and space debris accumulation. CubeSats, due to their low development cost and short mission timelines, are widely used for scientific, educational, and commercial applications. However, many CubeSat missions lack reliable end-of-life disposal mechanisms, leading to an increasing population of inactive spacecraft in LEO. International debris mitigation guidelines therefore emphasize the importance of ensuring timely deorbit of satellites after mission completion.

Active deorbiting using onboard propulsion systems provides a controlled and predictable method for orbital decay compared to passive techniques such as atmospheric drag augmentation. Traditional chemical propulsion systems can deliver high thrust but require relatively large amounts of propellant mass, which is a critical limitation for mass- and volume-constrained CubeSat platforms. In contrast, electric propulsion systems offer much higher specific impulse and consequently lower propellant mass requirements, although they operate with lower thrust levels and require longer maneuver durations. In recent years, green propulsion technologies have also emerged as safer and environmentally friendly alternatives to conventional chemical propellants for small spacecraft missions.

A first-order analytical treatment of this problem — comparing propellant mass requirements across propulsion types using the Tsiolkovsky rocket equation — provides useful early-stage design insight, but it necessarily neglects two physical effects that matter for a complete picture of CubeSat deorbit behavior: atmospheric drag, which acts continuously on any LEO spacecraft regardless of propulsion state, and the variability of atmospheric density with solar activity, which governs how strong that drag actually is at a given time. Deorbit is rarely a purely propulsion-driven process in practice — drag contributes throughout a spacecraft's orbital lifetime, and its strength depends on where that lifetime falls within the roughly 11-year solar cycle.

This study therefore proceeds in two stages. First, Sections 3.1 and 4.1 present a systems-level analytical trade study of green and electric propulsion for CubeSat deorbit, following the

approach of comparing propellant mass requirements via the Tsiolkovsky rocket equation, and examining how these requirements scale with spacecraft dry mass and available onboard power. Second, Sections 3.2 and 4.2 extend this baseline with a numerical orbit propagation framework — built using scipy.integrate.solve_ivp, with atmospheric density obtained from the NRLMSISE-00 model — that incorporates aerodynamic drag directly into the deorbit model, examines the coupled interaction between low-thrust propulsion and drag, and quantifies how solar activity, via its effect on atmospheric density, changes the rate of orbital decay. This extended methodology was developed in response to detailed technical feedback from Dr. Dario Modenini (University of Bologna, u3S Laboratory).

Together, these two stages provide a more complete picture of CubeSat deorbit behavior than either could alone: the analytical stage establishes propellant and power requirements under idealized assumptions, while the numerical stage tests how those requirements interact with two real physical effects — drag and solar-driven atmospheric variability — that are unavoidable in practice.

2. Mission Description and Assumptions

This study considers a representative 6U CubeSat mission operating in low Earth orbit (LEO) with the objective of performing a controlled deorbit at the end of its operational lifetime. The mission scenario is selected to be simple and general in order to allow comparison between different propulsion technologies under consistent conditions, and to support extension to a numerical propagation framework examining drag and solar activity effects under the same baseline conditions. The CubeSat is assumed to be initially deployed into a circular orbit and subsequently commanded to perform a deorbit maneuver to reduce its orbital altitude until atmospheric drag leads to natural reentry.

The baseline orbital parameters assumed in this study consist of an initial circular orbit altitude of 500 km and a target final altitude of 120 km. These values represent a typical operational LEO regime for CubeSat missions and a commonly adopted threshold for initiating rapid orbital decay. In addition, a reference spacecraft dry mass of 12 kg is adopted as the baseline CubeSat configuration for this study.

2.1: Analytical Modeling Assumptions

The Earth is modeled as a spherical body with standard gravitational parameters, and perturbative effects such as atmospheric drag during the maneuver, Earth oblateness, and third-body influences are neglected in order to focus on first-order propulsion performance trends.

A dry mass range of 5 kg to 25 kg is considered in the sensitivity analysis to represent different CubeSat class configurations and payload capacities. This range allows the investigation of how spacecraft mass scaling affects propellant requirements and propulsion system selection.

The spacecraft is assumed to carry a single propulsion system dedicated to the deorbit maneuver, and all propellant is used exclusively for end-of-life disposal.

Two propulsion technologies are considered in this study: green chemical propulsion and electric propulsion. The green propulsion system is characterized by a moderate specific impulse and relatively high thrust, while the electric propulsion system is characterized by a high specific impulse and low thrust. Representative specific impulse values are selected from typical literature ranges for small satellite propulsion systems. These parameters are used consistently across all analyses to ensure a fair comparison between propulsion options.

Several simplifying assumptions are made in order to maintain analytical tractability. The orbit is assumed to remain circular during the deorbit maneuver, and the required delta-V is computed using an idealized orbital velocity model. The propulsion systems are assumed to operate at constant performance throughout the maneuver, and losses due to inefficiencies, attitude control, and power system limitations are initially neglected. These assumptions enable a clear comparison of mass and performance trends while providing a baseline framework that can be extended in future work to include more detailed dynamics and system-level constraints.

This mission definition and set of assumptions provide a controlled environment for evaluating the impact of propulsion system selection on CubeSat deorbit performance. The framework established in this section supports subsequent analyses, including propellant mass comparison, sensitivity analysis with respect to CubeSat mass, and the incorporation of mission constraints such as deorbit time and available onboard power.

2.2: Extended Numerical Modeling Assumptions

The numerical simulations presented in Sections 3.2 and 4.2 retain the baseline mission parameters above (500 km initial altitude, 12 kg dry mass, Earth as central body) but relax several of the original simplifying assumptions in order to model orbital decay dynamically rather than through closed-form delta-V estimates alone.

The orbit is modeled as circular and sun-synchronous, with an inclination of 98°, consistent with real CubeSat missions at comparable altitudes (e.g., Aalto-1 at 97.4° and UWE-2 at 98.28°). Propagation is performed in the Earth-Centered Inertial (ECI) frame, required so that gravitational and drag accelerations can be computed without introducing fictitious rotating-frame forces; position is converted to the Earth-Centered Earth-Fixed (ECEF) frame and then to geodetic latitude, longitude, and altitude — using the WGS84 ellipsoid via Bowring's iterative method — solely to obtain atmospheric density at the satellite's true geodetic position at each timestep. Earth's gravity is modeled using the standard spherical two-body approximation; the J2 perturbation, which primarily affects orbital plane orientation (nodal and apsidal precession) rather than altitude decay, is not included, since altitude decay is the focus of this study. Atmospheric drag is computed relative to the co-rotating atmosphere, assumed to rotate rigidly with Earth.

A reentry condition is defined at 120 km altitude, consistent with the target altitude used in the baseline analytical study (Table 1); numerical integration is terminated when this altitude is reached, since the two-body-plus-drag model is no longer physically meaningful below it.

3. Methodology

3.1: Analytical Modeling Methodology

The methodology adopted in this study combines analytical orbital mechanics with a computational simulation framework developed in Python. The objective is to quantify the propellant mass required for CubeSat deorbit maneuvers using different propulsion technologies and to evaluate the sensitivity of these requirements to variations in spacecraft mass and mission parameters.

The required delta-V for deorbit is calculated based on the change in orbital velocity between the initial and final circular orbits using a Hohmann transfer approximation. Assuming a circular orbit at altitude h1 and a target orbit at altitude h2, the orbital velocity is computed using the standard gravitational parameter of Earth and the orbital radius. The delta-V is obtained from the difference between the two orbital velocities. This approach provides a first-order estimate of the velocity change required to initiate orbital decay.

The propellant mass required for the maneuver is calculated using the Tsiolkovsky rocket equation:
Delta V = Isp g0 ln (m0 / mf)
where Isp is the specific impulse of the propulsion system, g0 is the standard gravitational acceleration, m0 is the initial spacecraft mass before the maneuver, and mf is the final mass after propellant expenditure. Rearranging this equation yields the propellant mass required for a given delta-V and spacecraft mass.

Representative propulsion parameters are selected for two propulsion technologies: green chemical propulsion and electric propulsion. Each system is assigned a constant specific impulse value based on typical performance ranges reported in small satellite propulsion literature. These parameters are used consistently across all simulation cases to ensure comparability between propulsion options.

A modular Python code structure is employed to implement the analytical model. Core functions are developed to compute orbital velocity, delta-V, and propellant mass. These functions are reused across multiple analysis scripts, including baseline propulsion comparison and sensitivity analysis. Input parameters such as CubeSat mass, propulsion system type, and orbital altitudes are defined at the top of each script to allow rapid modification and parametric studies.

For the sensitivity analysis, the CubeSat dry mass is varied over a defined range while holding orbital parameters and propulsion characteristics constant. For each mass value, the delta-V and corresponding propellant mass are computed for both propulsion systems. The results are stored in structured data containers and visualized using bar plots and line plots generated with the Matplotlib library.

All simulations are executed in a Visual Studio Code (VS Code) environment to enable interactive testing, verification, and visualization. The generated plots are saved as image files and incorporated into the technical report for interpretation and discussion. This computational methodology enables reproducible analysis and provides a foundation for extending the study to include additional constraints such as power availability and deorbit time.

3.2: Numerical Simulation Methodology

To extend the analytical results beyond idealized, instantaneous-maneuver assumptions, a numerical orbit propagation framework was developed in Python, using scipy.integrate.solve_ivp to integrate the CubeSat's equations of motion forward in time under gravity, atmospheric drag, and, where applicable, propulsive thrust.

3.2.1: Reference Frames and Coordinate Transformations

Propagation is performed in the Earth-Centered Inertial (ECI) frame. Initial position and velocity are computed in the perifocal (PQW) frame from the classical orbital elements (Section 2.2) and rotated into ECI via the standard 3-1-3 Euler sequence (right ascension of the ascending node, inclination, argument of perigee). At each integration step, position is additionally converted from ECI to the Earth-Centered Earth-Fixed (ECEF) frame, using Earth's rotation angle computed from the sidereal rotation period, and then to geodetic latitude, longitude, and altitude via the WGS84 ellipsoid, using Bowring's iterative method. This conversion is used solely to obtain the satellite's true geodetic position for the atmospheric density lookup described below; it does not affect the equations of motion themselves, which are evaluated entirely in the inertial ECI frame.

3.2.2: Gravitational and Drag Accelerations

Gravitational acceleration is modeled using the standard spherical two-body approximation. Atmospheric density is obtained from the NRLMSISE-00 empirical atmospheric model (via the pymsis Python package), queried at the satellite's current geodetic position and simulation date. Drag acceleration is computed from the standard drag equation,

$F_{drag} = \frac{1}{2} \rho C_d A v_{rel}^2$

where ρ is atmospheric density, $C_d = 2.2$ is the drag coefficient (a standard value for CubeSat-class objects in free molecular flow), $A = 0.02$ m² is the spacecraft's cross-sectional area (smallest face of a 6U CubeSat, nose-first), and v_{rel} is the satellite's velocity relative to the atmosphere. The atmosphere is assumed to co-rotate rigidly with Earth; v_{rel} is therefore computed as the satellite's inertial velocity minus the local atmospheric velocity due to Earth's rotation.

3.2.3: Reentry Condition

Integration is terminated once altitude reaches 120 km (REENTRY_ALTITUDE_KM), matching the target altitude of the baseline analytical study (Table 1), implemented as a terminal event condition in solve_ivp. Below this altitude, the two-body-plus-drag model no longer represents the physical regime of the spacecraft, and continued integration produces non-physical results.

3.2.4: Propulsion Modeling

Where thrust is included, it is applied retrograde (opposite the instantaneous ECI velocity vector) continuously over a fixed burn window, and propellant mass is tracked as an additional state variable, consumed according to the rocket thrust equation, $F = v_e (dm/dt)$, where $v_e = I_{sp} g_0$ is effective exhaust velocity. Current total spacecraft mass (dry mass plus remaining propellant) is used in both the thrust and drag acceleration calculations, coupling the two effects: as propellant is expended, both available thrust-to-mass ratio and drag deceleration per unit force change together.

4. Results and Discussion

4.1: Analytical Results and Discussion

Section 4.1 presents the outputs obtained from the developed Python-based simulation framework. The results include a comparison of propulsion strategies for CubeSat deorbiting and a series of parametric trade studies. These trade studies investigate the influence of spacecraft mass, power and propulsion system characteristics, and deorbit time constraints on the required propellant mass and overall system performance.

4.1.1: Propellant Mass Comparison for CubeSat Deorbit

Section 4.1.1 presents the results of the analytical deorbit model and compares the propellant mass requirements for different propulsion systems. The analysis considers a CubeSat deorbit

maneuver from an initial circular orbit at an altitude of 500 km to a target altitude of 120 km. The spacecraft dry mass is assumed to be 12 kg, and Earth is taken as the central body. These mission parameters are summarized in Table 1.

Parameter	Value
Spacecraft configuration	6U CubeSat
Initial orbit altitude	500 km
Target orbit altitude	120 km
Spacecraft dry mass	12 kg
Central body	Earth

Table 1: Mission parameters used for the deorbit analysis.

The required velocity change (delta-V) for the deorbit maneuver was calculated using the analytical orbital mechanics model described in Section 3.1. The resulting delta-V requirement is presented in Table 2.

Quantity	Value
Required deorbit delta-V	109.08 m/s

Table 2: Required delta-V for the deorbit maneuver.

Using the computed delta-V value, the propellant mass required for two different propulsion systems — green chemical propulsion and electric propulsion — was determined using the Tsiolkovsky rocket equation. The specific impulse values assumed for the propulsion systems were 220 s for green propulsion and 1200 s for electric propulsion. The resulting propellant mass requirements are summarized in Table 3.

Propulsion system	Specific impulse (s)	Required propellant mass (kg)
Green propulsion	220	*0.6223*
Electric propulsion	1200	*0.1118*

Table 3: Comparison of propellant mass requirements for different propulsion systems.

The comparison of propellant mass requirements for the two propulsion systems is illustrated in Figure 1.

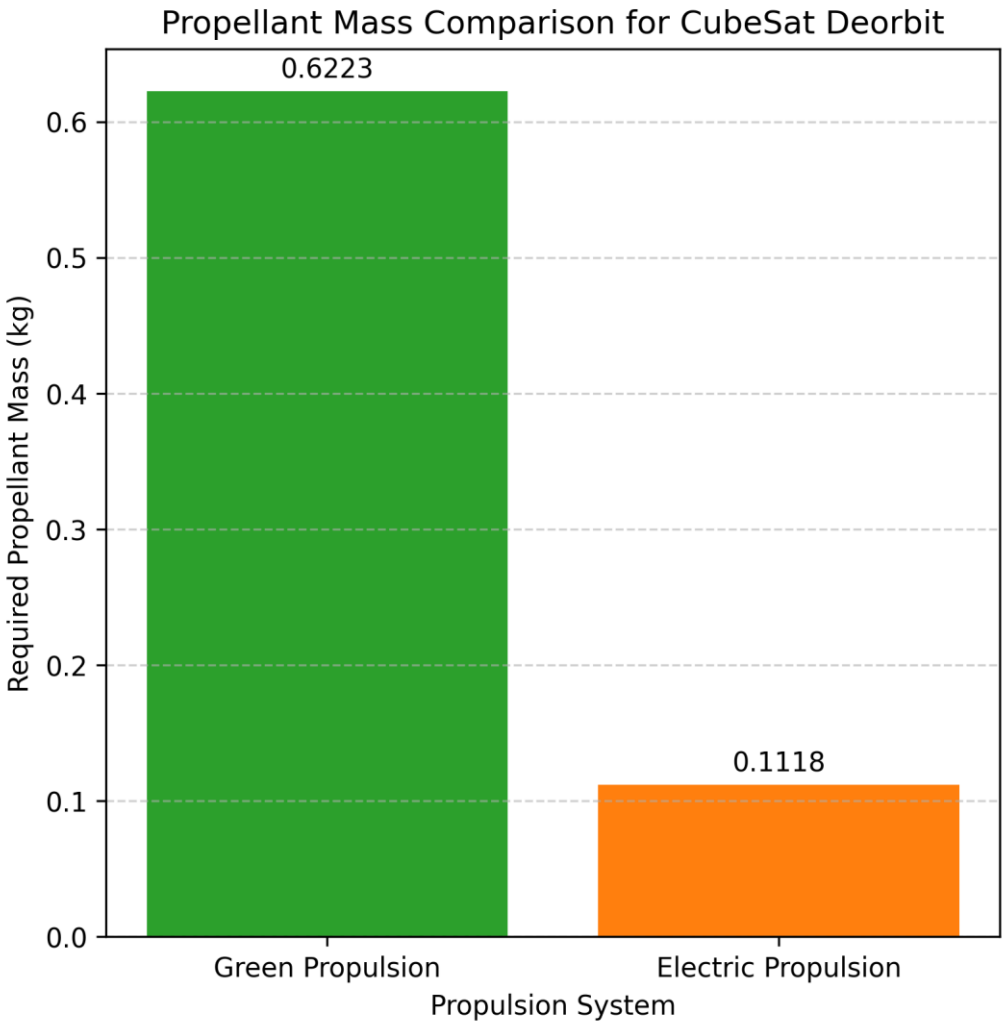


Figure 1: Comparison of required propellant mass for green and electric propulsion systems for CubeSat deorbit.

For a deorbit maneuver from an initial altitude of 500 km to a target altitude of 120 km, a total delta-V of 109.08 m/s is required. Based on this delta-V, green propulsion requires approximately 0.62 kg of propellant, whereas electric propulsion requires only about 0.11 kg. This significant reduction in propellant mass highlights the advantage of high specific impulse propulsion systems for CubeSat deorbit missions.

Electric propulsion systems enable substantial mass savings, which can be utilized for additional payload capacity, enhanced subsystem margins, or extended mission duration. These results demonstrate that propulsion system selection plays a critical role in the overall mass budget of CubeSat deorbit strategies and motivates further investigation into system-level trade-offs involving mass, power, and deorbit time.

4.1.2: Sensitivity Analysis of Propellant Mass with Respect to CubeSat Dry Mass

This subsection investigates the sensitivity of the required propellant mass to variations in CubeSat dry mass. While the previous analysis considered a fixed spacecraft dry mass of 12 kg, practical CubeSat missions span a wide range of masses depending on payload, subsystem configuration, and form factor. Therefore, a parametric study was performed by varying the CubeSat dry mass and evaluating its effect on the required propellant mass for different propulsion systems.

In this analysis, the CubeSat dry mass was varied over a representative range while keeping all other mission parameters constant, including the initial orbit altitude of 500 km, the target orbit altitude of 120 km, and the required deorbit delta-V of 109.08 m/s. The same propulsion system characteristics used in Section 4.1.1 were retained, with specific impulse values of 220 s for green propulsion and 1200 s for electric propulsion.

For each selected CubeSat dry mass value, the corresponding propellant mass was computed using the Tsiolkovsky rocket equation. This approach allows direct quantification of how spacecraft mass scaling impacts the propellant budget for deorbit maneuvers.

The results of the sensitivity analysis are presented in Figure 2, which shows the variation of required propellant mass as a function of CubeSat dry mass for both propulsion systems.
The corresponding numerical values of propellant mass for each CubeSat dry mass are summarized in Table 4.

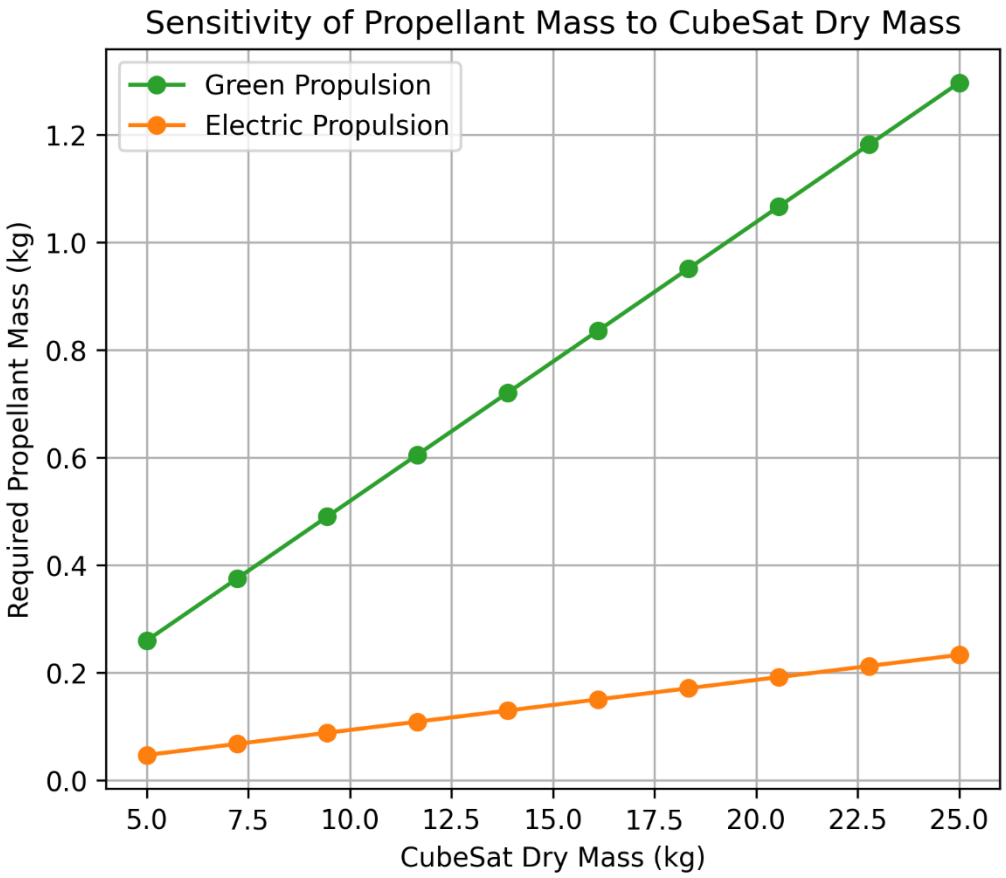


Figure 2: Sensitivity of required propellant mass to CubeSat dry mass for green and electric propulsion systems.

CubeSat Dry Mass (kg)	Propellant Mass – Green Propulsion (kg)	Propellant Mass – Electric Propulsion (kg)
5.0	0.2593	0.0466
7.2	0.3746	0.0673
9.4	0.4898	0.0880
11.7	0.6050	0.1086
13.9	0.7203	0.1293
16.1	0.8355	0.1500
18.3	0.9508	0.1707
20.6	1.0660	0.1914
22.8	1.1813	0.2121
25.0	1.2965	0.2328

Table 4: Required propellant mass as a function of CubeSat dry mass for green and electric propulsion systems.

The results indicate that the required propellant mass increases monotonically with increasing CubeSat dry mass for both propulsion systems. However, the rate of increase differs significantly between the two cases. Green propulsion exhibits a steeper growth in propellant mass as spacecraft mass increases, whereas electric propulsion shows a comparatively gradual increase due to its much higher specific impulse.

For lower-mass CubeSats, the propellant mass required by green propulsion remains relatively small; however, as spacecraft mass increases, the propellant fraction becomes a more substantial portion of the total system mass. In contrast, electric propulsion maintains low propellant mass requirements even for higher spacecraft dry masses, making it more scalable for larger CubeSat platforms.

This sensitivity analysis highlights the strong dependence of deorbit propellant requirements on spacecraft mass and demonstrates the importance of considering mass scaling effects during early mission design. The results further emphasize the benefits of high specific impulse propulsion systems in minimizing propellant mass penalties, particularly for CubeSat missions with increasing payload or subsystem mass.

Overall, the sensitivity study provides quantitative insight into the trade-offs between spacecraft mass and propulsion system selection and forms a basis for system-level optimization when designing CubeSat deorbit strategies.

4.1.3: Power Constraint and Propulsive Burn Time Trade Study

This section presents a combined power constraint analysis and propulsive burn time trade study for the CubeSat deorbit mission using electric propulsion. The power constraint analysis evaluates the thrust capability of the propulsion system under different available onboard power levels, while the propulsive burn time trade study examines the duration of thrusting required to achieve the computed deorbit delta-V. Together, these analyses assess the feasibility of electric propulsion under realistic CubeSat power limitations.

For a fixed mission delta-V and spacecraft mass, the thrust generated by an electric propulsion system is fundamentally limited by the available electrical power. Lower power levels result in reduced thrust and mass flow rate, thereby increasing the total propulsive burn time required to complete the deorbit maneuver. Conversely, higher power availability enables greater thrust and shorter burn durations. This trade-off is critical for CubeSat missions, where onboard power generation capability is constrained by spacecraft size and solar array area.

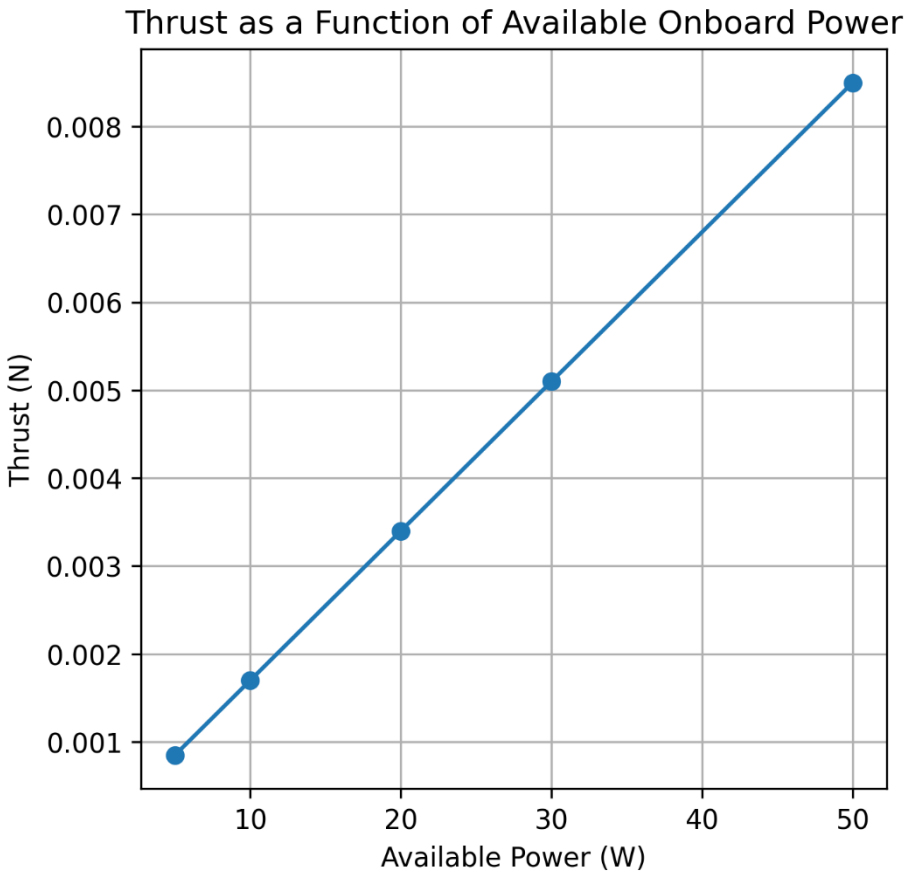


Figure 3: Thrust as a function of available onboard power for electric propulsion.

Figure 3 illustrates the dependence of propulsion thrust on available electrical power. The results indicate a direct relationship between power and thrust, confirming that electric propulsion performance is strongly constrained by the CubeSat power subsystem. At low power levels, the generated thrust is minimal, which limits maneuver aggressiveness and extends mission timelines. As available power increases, thrust capability improves, enhancing maneuver feasibility but requiring larger power generation and storage systems.

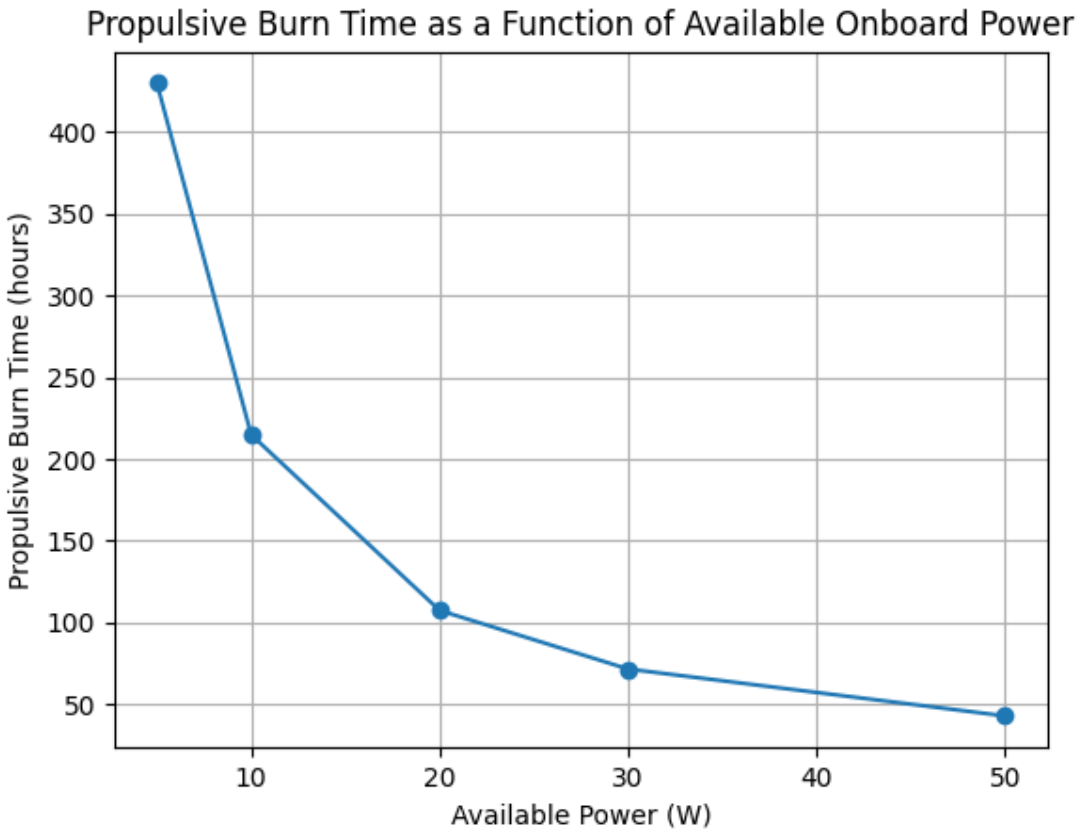


Figure 4: Propulsive burn time as a function of available onboard power for electric propulsion.

Figure 4 presents the variation of propulsive burn time with available onboard power. The results show that burn time decreases significantly as power increases, reflecting the higher thrust and mass flow rate achievable at higher power levels. At low power levels, the required burn duration becomes excessively long, which may be impractical due to operational constraints, attitude control limitations, and mission lifetime considerations. These findings highlight the importance of balancing propulsion capability with realistic power budgets in CubeSat design.

Available Power (W)	**Thrust (N)**	**Propulsive Burn Time (hours)**
5	0.000850	429.89
10	0.001700	214.94
20	0.003399	107.47
30	0.005099	71.65
50	0.008498	42.99

Table 5: Power constraint and propulsive burn time trade study results.

Table 5 summarizes the numerical results of the power constraint and burn time analyses. The table provides quantitative insight into how increasing power levels improve thrust capability while reducing the required propulsive burn duration. These results reinforce the trends observed in Figures 3 and 4 and enable direct comparison across discrete power scenarios. Such tabulated data is useful for preliminary spacecraft design decisions, allowing designers to evaluate feasible combinations of propulsion performance and power system capacity.

The combined power constraint and propulsive burn time trade study demonstrates that electric propulsion feasibility for CubeSat deorbit missions is governed primarily by available onboard power. While electric propulsion offers propellant mass advantages, its low-thrust nature results in extended burn durations under limited power conditions. These findings emphasize the need for integrated design of propulsion and power subsystems and provide a quantitative framework for assessing trade-offs in small satellite deorbit mission planning.

4.2: Numerical Simulation Results and Discussion

4.2.1: Aerodynamic Drag Modeling

The first extension examined how atmospheric drag alone, with no propulsion, affects the CubeSat's orbital altitude over time. Starting from the baseline 500 km circular, sun-synchronous orbit (Section 2.2), the spacecraft was propagated for 365 days under gravity and drag only.

Over this period, altitude decreased from 500.000 km to 497.034 km, a total decay of 2.966 km (Figure 5). This magnitude of decay is consistent with published CubeSat decay-time studies, which report decay periods on the order of 10–15 years for comparable spacecraft at 500 km altitude under average solar conditions — a slow initial decay rate is expected at this altitude, since drag increases substantially only as altitude decreases. The decay curve itself shows a small periodic oscillation superimposed on the overall downward trend, attributable to the orbital period (approximately 94.6 minutes at 500 km) being under-sampled by the fixed output resolution used for plotting; the underlying integration itself uses adaptive, much finer timesteps and is not affected by this sampling artifact.

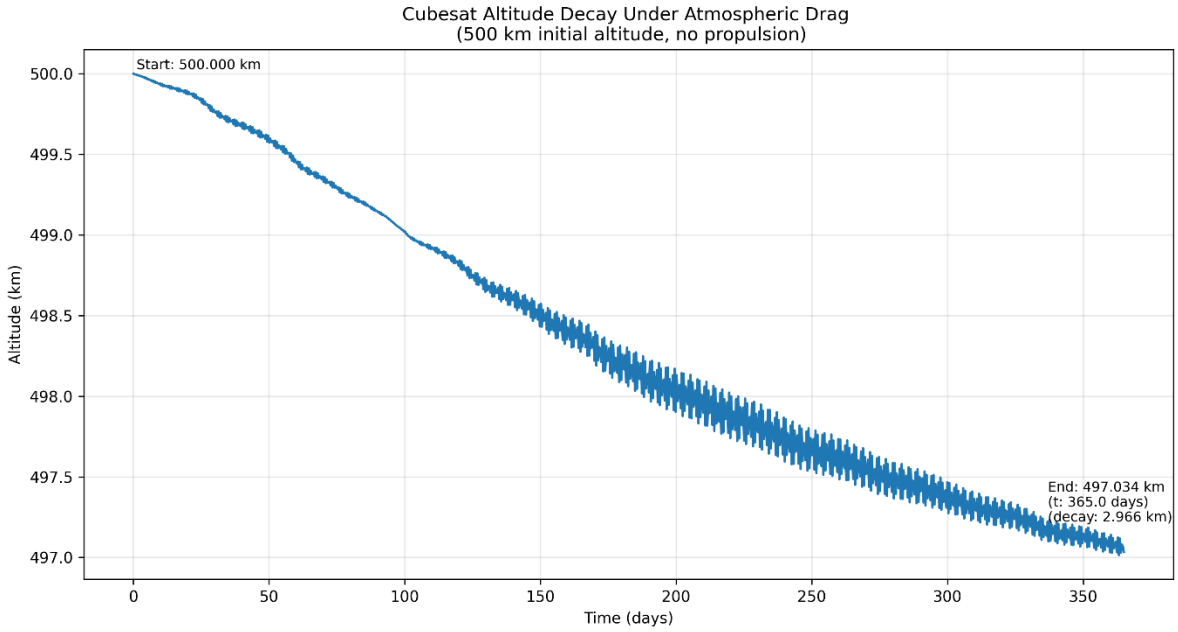


Figure 5: Altitude decay under atmospheric drag alone, 500 km initial altitude, 365-day propagation.

4.2.2: Coupled Propulsion and Drag Interaction

The second extension examined deorbit scenarios in which propulsive thrust and atmospheric drag act together, rather than treating deorbit as a purely propulsion-driven process. The propellant masses computed analytically in Section 4.1 (0.1118 kg for electric propulsion, 0.6223 kg for green propulsion, both sized for the idealized 500 km → 120 km Hohmann-style transfer of Table 3) were applied here as fixed propellant budgets, consumed instead as a continuous retrograde burn, with real gravity and drag acting throughout. This represents a deliberate shift from the preliminary analytical sizing of Section 4.1 to a higher-fidelity dynamic simulation: the numerical results below should be read as examining how the same propellant allowance performs under continuous-burn dynamics with drag present, not as a reproduction of the idealized instantaneous transfer.

Electric propulsion thrust was taken as 0.005099 N, corresponding to the 30 W operating point of the power-thrust relationship established in Section 4.1.3 (Table 5); green propulsion thrust was taken as 0.5 N, representative of a small CubeSat monopropellant thruster (CU Aerospace MPUC/CMP-X), which demonstrated thrust exceeding 0.5 N in ground testing.

Both scenarios were propagated until reentry (120 km altitude) or 365 days, whichever occurred first. The electric propulsion scenario reached reentry at 163.7 days; the green propulsion scenario reached reentry at 62.1 days (Figure 6). Both scenarios reached reentry well within the simulated year, confirming that the propellant budgets established analytically in Section 4.1.1, when expended continuously rather than instantaneously, are sufficient to

drive the spacecraft to reentry altitude even with drag acting as an additional, independent decelerating force throughout.

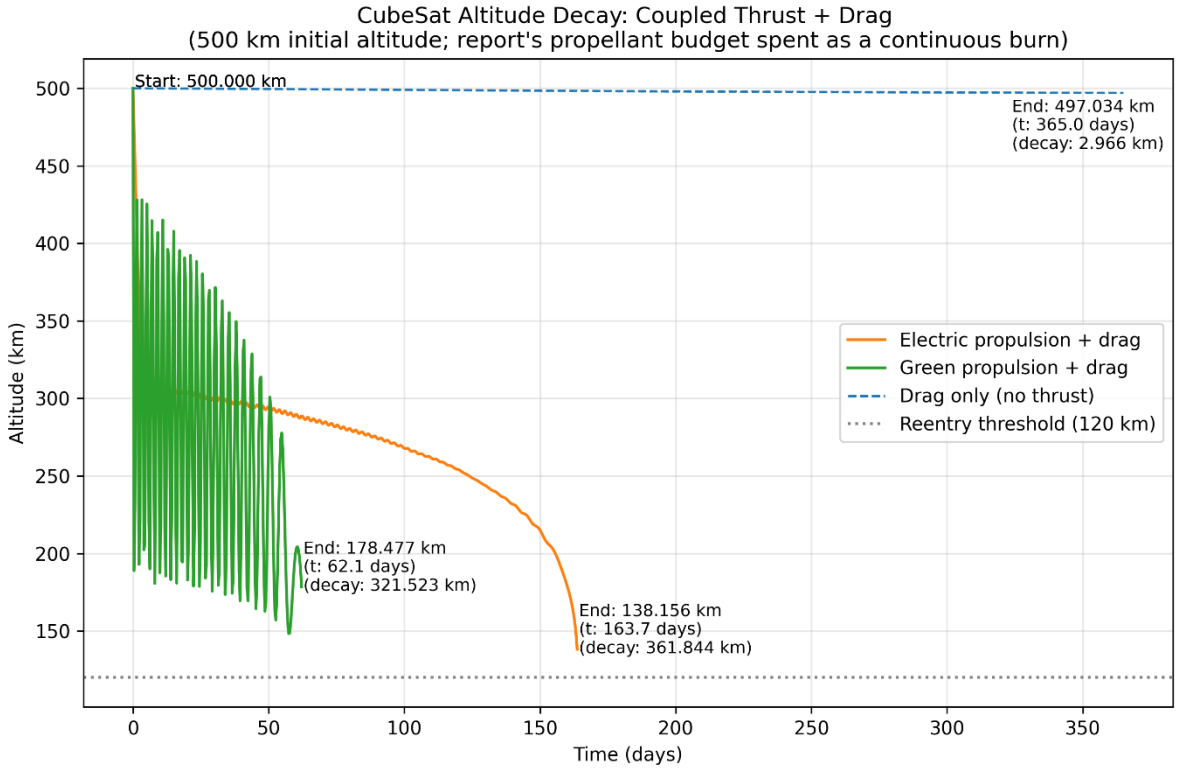


Figure 6: Altitude decay under coupled thrust and drag, electric and green propulsion scenarios, compared against the drag-only baseline.

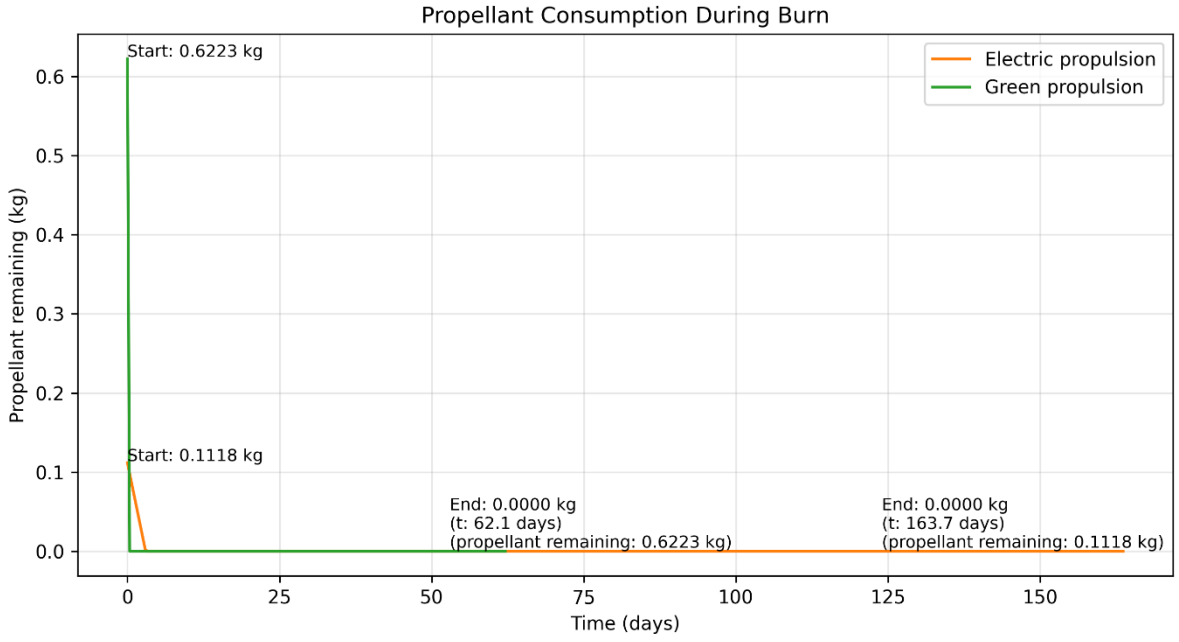


Figure 7: Propellant mass remaining over time for electric and green propulsion during the coupled thrust-and-drag burn.

Figure 7 shows that the propellant mass decreases to zero in both propulsion scenarios, with the green propulsion system exhausting its propellant sooner than the electric propulsion system. This reflects the shorter burn duration of the green propulsion system under the assumed operating conditions.

To examine how drag's relative contribution varies with thrust regime, as specifically requested in Dr. Dario Modenini's feedback, the electric propulsion scenario was additionally run at all five power levels established in Section 4.1.3 (Table 5: 5, 10, 20, 30, and 50 W), each using the same 0.1118 kg propellant budget, consumed over a correspondingly different burn duration.

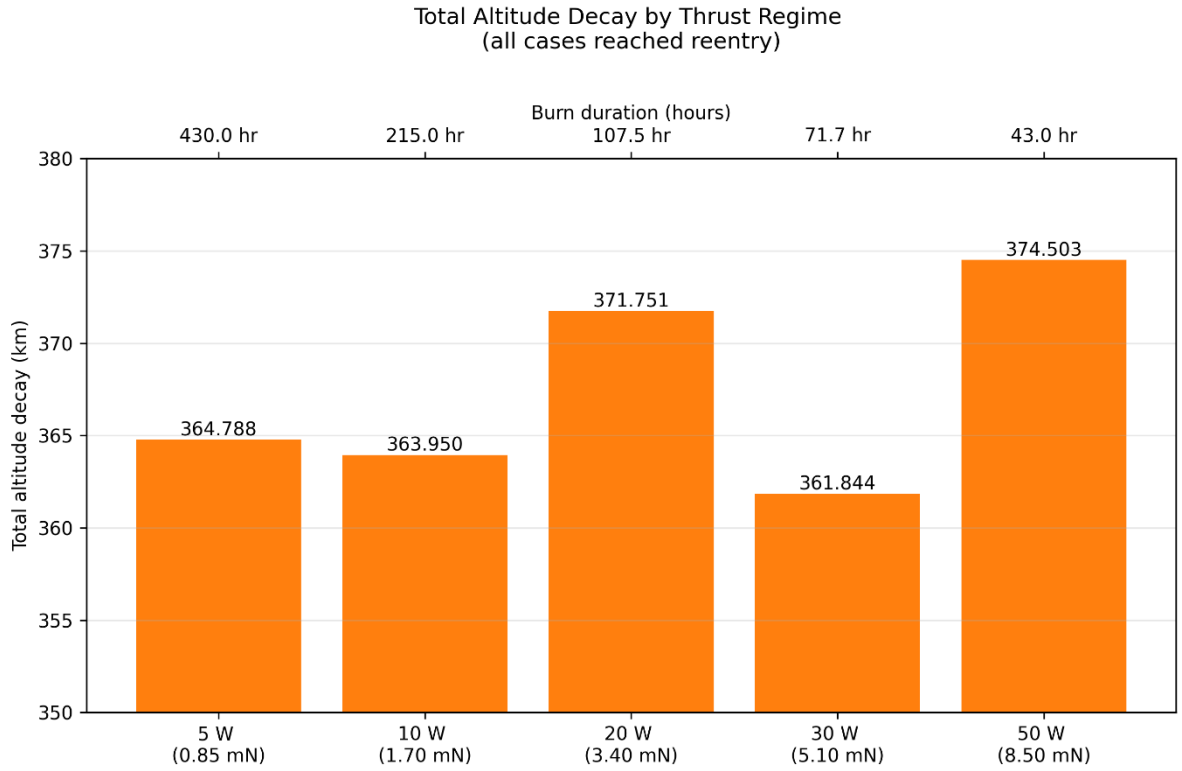


Figure 8: Total altitude decay at reentry across the five electric propulsion thrust regimes of Table 5 (5 – 50 W), shown alongside their corresponding thrust values.

Figure 8 presents this same total-decay data as a bar chart; while total decay varies modestly across regimes (361.8 – 374.5 km), this variation does not track monotonically with thrust level, consistent with drag's contribution remaining small and roughly constant regardless of burn duration.

Across all five regimes, total altitude decay at reentry ranged from 361.8 km to 374.5 km, and drag's contribution — isolated by comparison against a drag-only baseline propagated over the same elapsed time — remained small and roughly constant, between 0.43% and 0.48% of total decay, across the full power range (Figure 9, Table 6). This indicates that, for the propellant budget and thrust levels considered here, propulsive thrust dominates total altitude change at every tested power level; drag's relative contribution does not vary substantially with burn duration within this regime, since even the longest burn (429.95 hours, at 5 W) remains short relative to the multi-year timescale over which drag alone would produce comparable decay (Section 4.2.1)

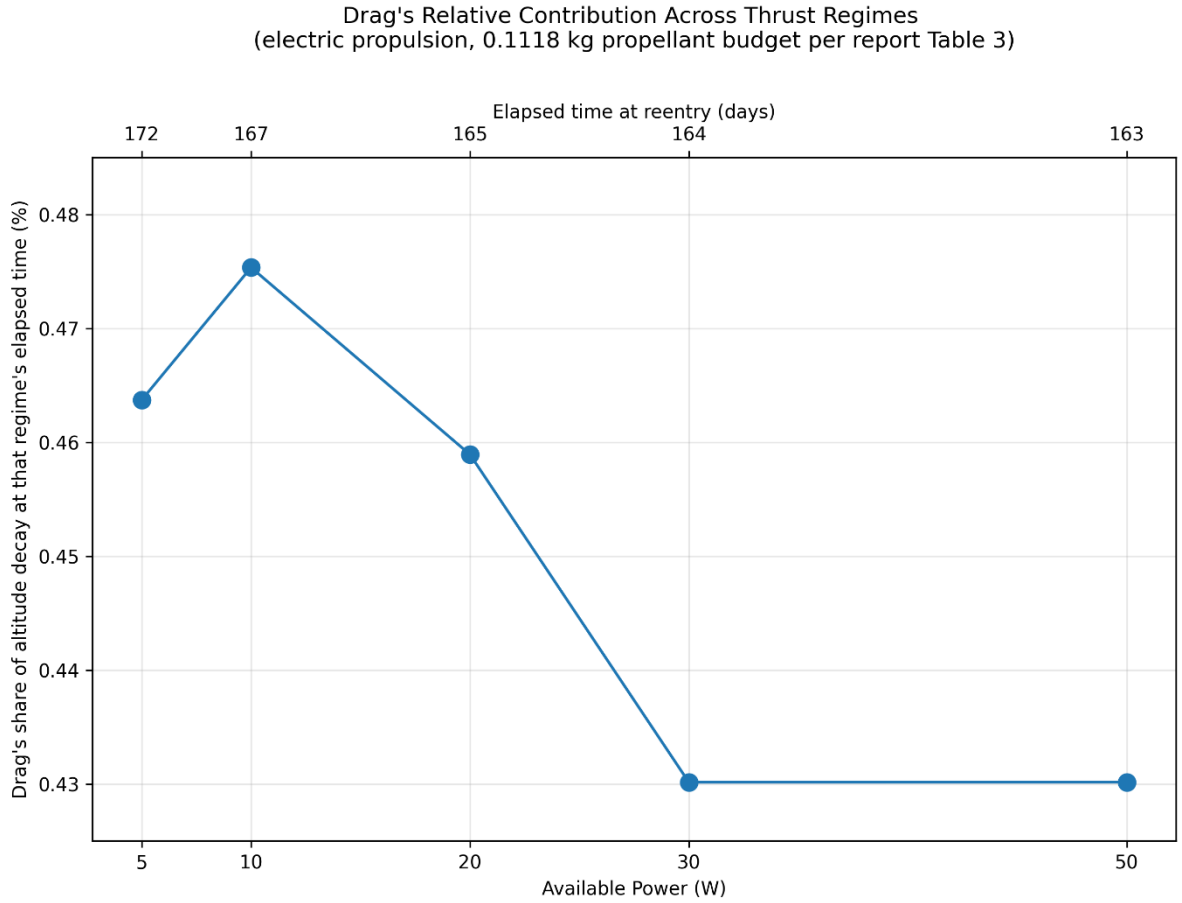


Figure 9: Drag's percentage contribution to total altitude decay across five electric propulsion thrust regimes.

Power (W)	Thrust (mN)	Burn Duration (hours)	Elapsed Time to Reentry (days)	Total Decay (km)	Drag's Share (%)
5	0.85	429.95	172.09	364.788	0.46
10	1.70	214.98	167.34	363.950	0.48
20	3.40	107.52	164.78	371.751	0.46
30	5.10	71.67	163.68	361.844	0.43

50	8.50	43.01	163.32	374.503	0.43

Table 6: Total altitude decay and drag's relative contribution by thrust regime.

4.2.3: Sensitivity to Solar Activity

The third extension examined how solar activity, via its effect on atmospheric density, affects drag-driven orbital decay. The drag-only simulation of Section 4.2.1 was repeated with four different simulation start dates, each corresponding to a distinct phase of Solar Cycle 25: solar minimum (December 2019, smoothed sunspot number = 1.8, F10.7 ≈ 68 SFU), rising phase (April 2022, F10.7 ≈ 116 SFU), near-peak (December 2022, F10.7 ≈ 142 SFU), and solar maximum (August 2024, smoothed sunspot number = 156.8, F10.7 ≈ 245.6 SFU — the single highest monthly value of Solar Cycle 25, confirmed directly from NOAA's Solar Cycle Progression data). Each scenario was propagated for 365 days; because the simulation date advances continuously throughout each propagation, with atmospheric density queried from NRLMSISE-00 at the satellite's actual simulated date rather than held fixed at the starting value, each scenario's label describes its starting solar activity condition, which — given the timescales involved relative to the ~11-year solar cycle — remains broadly representative of that condition throughout the one-year propagation window.

Total altitude decay over 365 days increased monotonically with solar activity: 0.52 km at solar minimum, rising to 3.52 km in the rising phase, 4.73 km near-peak, and 7.04 km at solar maximum (Figure 10, Table 7). The simulated decay at solar maximum was approximately 13.57 times that at solar minimum. This result directly demonstrates the physical mechanism underlying solar-driven drag variability: elevated solar activity heats and expands Earth's thermosphere, increasing atmospheric density at a given altitude and correspondingly increasing drag force on an orbiting spacecraft.

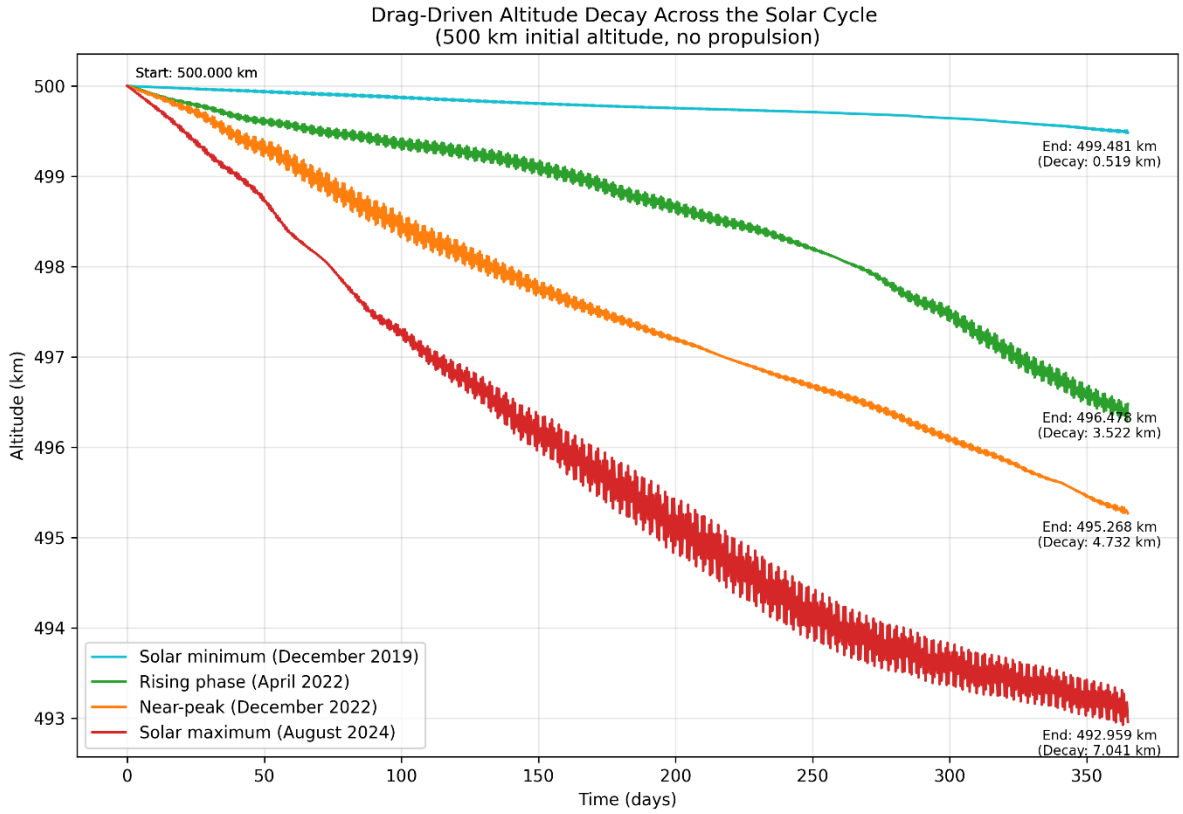


Figure 10: Drag-driven altitude decay across four phases of Solar Cycle 25.

Scenario	Date	Solar Activity Indicator	Total Decay over 365 days (km)
Solar minimum	December 2019	F10.7 ≈ 68 SFU	0.52
Rising phase	April 2022	F10.7 ≈ 116 SFU	3.52
Near-peak	December 2022	F10.7 ≈ 142 SFU	4.73
Solar maximum	August 2024	F10.7 ≈ 245.6 SFU	7.04

Table 7: Total altitude decay by solar cycle phase.

To facilitate direct comparison of the total altitude decay across the four solar-activity scenarios, Figure 11 presents the results from Table 7 as a bar chart.

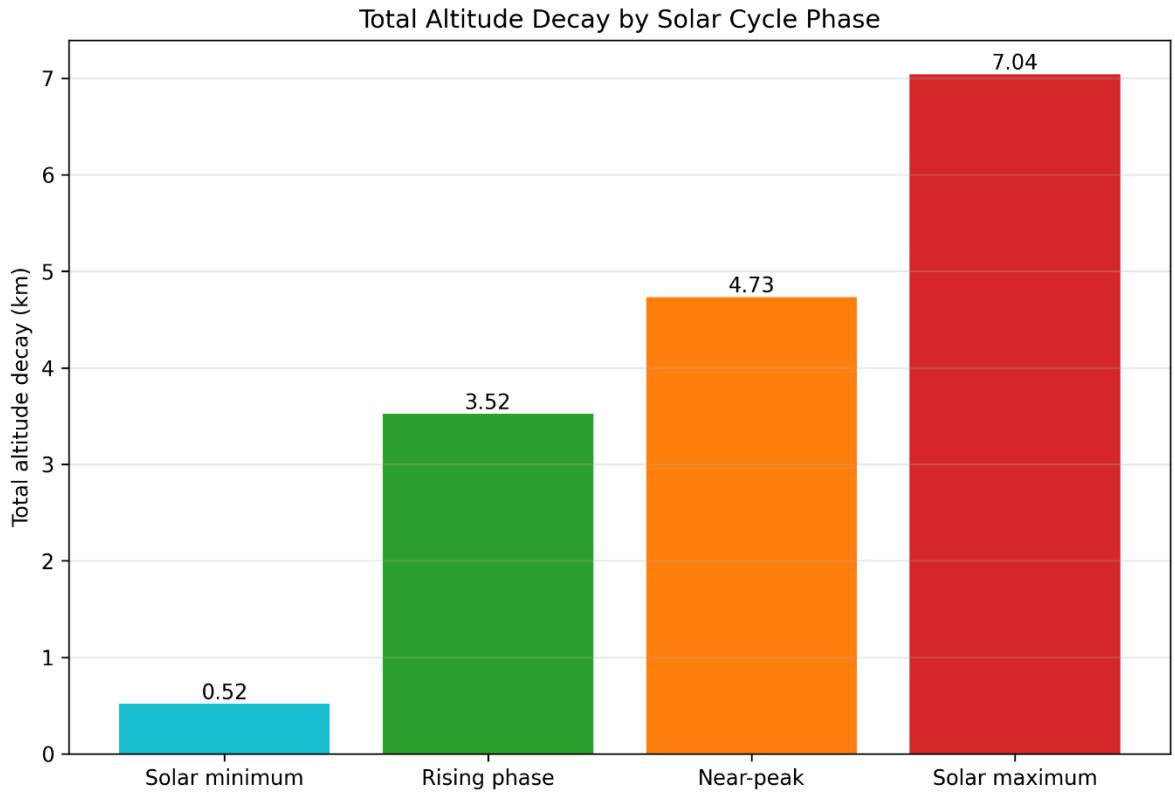


Figure 11: Total altitude decay over 365 days at each of the four Solar Cycle 25 phases examined, shown as a bar chart for direct comparison.

Figure 11 highlights the magnitude of the difference between the solar-activity scenarios, with the simulated altitude decay at solar maximum approximately 13.57 times that at solar minimum. This comparison demonstrates that solar activity can substantially influence long-term orbital decay, reinforcing the importance of accounting for atmospheric variability when predicting CubeSat orbital lifetimes.

4.3: Combined Discussion

The analytical and numerical stages of this study, taken together, provide a more complete picture of CubeSat deorbit behavior than either could alone. The analytical results of Section 4.1 established that electric propulsion offers substantial propellant mass savings over green propulsion (0.1118 kg versus 0.6223 kg, Table 3) for the baseline 500 km → 120 km transfer, at the cost of lower thrust and correspondingly longer burn durations — a trade off directly quantified by the power-constraint results of Section 4.1.3, where burn duration ranged from 42.99 hours at 50 W to 429.89 hours at 5 W (Table 5). These findings, derived from an idealized instantaneous-maneuver model, raise a natural question the analytical stage alone cannot answer: how do these propellant budgets and burn durations actually behave once continuous propulsion, atmospheric drag, and real orbital dynamics are all modeled together? Section 4.2 addresses this directly.

First, the drag-only results (Section 4.2.1) show that atmospheric drag alone produces altitude decay on a multi-year timescale at 500 km — far slower than any propulsion-driven maneuver considered in Section 4.1. This confirms, independently of the propulsion comparison, that active propulsion remains necessary for timely deorbit within a practical mission lifetime; drag augmentation alone would not satisfy typical debris-mitigation timelines from this altitude, reinforcing the premise underlying the entire analytical trade study.

Second, the coupled propulsion-drag results (Section 4.2.2) test whether the propellant budgets computed analytically in Section 4.1.1 actually suffice once applied realistically, as continuous burns with drag acting throughout, rather than as the instantaneous maneuvers the Tsiolkovsky-based sizing assumes. Both propulsion types reached reentry well within the simulated year — electric propulsion at 163.7 days, green propulsion at 62.1 days — confirming that the analytical propellant sizing of Section 4.1.1 remains valid even under this more realistic continuous-burn, drag-present model. The five-point thrust regime sweep (Table 6), built directly on the power-thrust relationship established in Section 4.1.3, further shows that propulsive thrust dominates total altitude change at every tested power level, with drag contributing a small, roughly constant fraction (0.43 – 0.48%) regardless of burn duration. This extends the Section 4.1.3 power-constraint finding — that lower power means longer burns — by showing that those longer burns do not substantially change drag's relative role in the outcome. For mission designs using propellant budgets of this scale, drag's contribution can therefore reasonably be treated as a secondary, relatively stable effect rather than one requiring detailed re-optimization across different thrust regimes — though this conclusion is specific to the propellant masses and thrust levels examined here and may not generalize to significantly longer burns or lower-mass CubeSats with a higher area-to-mass ratio, a scaling relationship the Section 4.1.2 sensitivity analysis suggests would be worth revisiting numerically in future work.

Third, the solar activity results (Section 4.2.3) add a dimension absent from both the analytical model and the drag-only and coupled numerical results above: time-dependence driven by the solar cycle. Even though drag's absolute contribution to total decay is small relative to propulsion in the coupled scenarios, drag-only decay itself varies by more than a factor of 13.57 between solar minimum and solar maximum conditions (0.52 km versus 7.04 km over one year, Table 7). This has a practical implication that the analytical stage, with its fixed, idealized assumptions, could not surface: a CubeSat relying partly on drag for end-of-life disposal, or one whose mission planning assumes a nominal atmospheric density, should account for where in the solar cycle its mission falls, since decay rate — and therefore uncompensated orbital lifetime — can differ substantially depending on solar conditions at the time of deorbit.

Considered together, the analytical and numerical results are consistent with, and mutually reinforcing of, one another: the propulsion comparison and propellant sizing of Section 4.1 hold up under the more realistic dynamics of Section 4.2, while the numerical stage adds exactly the physical effects — continuous-time drag, atmospheric variability — that a closed-

form delta-V model cannot represent. Several simplifications remain across the numerical extensions, stated explicitly here for completeness. The J2 gravitational perturbation is not included; its primary effect is on orbital plane orientation (nodal and apsidal precession) rather than altitude decay, and its omission is not expected to materially affect the altitude-decay conclusions presented here, though it would be necessary for any future extension examining ground-track or orbital-plane evolution. Earth is modeled as the WGS84 ellipsoid for the purpose of geodetic position and atmospheric density lookup — a refinement beyond what was explicitly requested, adopted for improved physical fidelity — but gravity itself retains the simpler spherical two-body approximation used throughout Section 4.1 as well. Atmospheric co-rotation is included in the drag calculation, also beyond what was explicitly requested, as a refinement to the basic drag equation.

5. Conclusion

This study presented a systems-level trade analysis of CubeSat deorbit strategies, combining an analytical propellant-mass and sensitivity study with a numerical orbit propagation framework incorporating atmospheric drag, coupled propulsion-drag dynamics, and solar activity sensitivity. The analytical results demonstrated that electric propulsion offers substantial propellant mass savings over green chemical propulsion, whereas green chemical propulsion's higher propellant requirements place greater demands on the spacecraft mass budget. However, electric propulsion's mass efficiency comes at the cost of lower thrust and correspondingly longer burn durations, with onboard power availability acting as the primary constraint on its feasibility. The sensitivity analysis further highlighted how increases in spacecraft dry mass lead to higher propellant demand, underscoring the importance of considering mass, power availability, and mission timeline together when selecting a deorbit strategy.

Extending this baseline numerically showed that atmospheric drag alone decays a 500 km CubeSat orbit only slowly, on a multi-year timescale, confirming the necessity of active propulsion for timely deorbit; that the propellant budgets established analytically remain sufficient to achieve reentry under realistic continuous-burn dynamics, with drag contributing a small and roughly constant share of total decay across the thrust regimes examined; and that solar activity meaningfully modulates drag-driven decay rate, with decay at solar maximum exceeding twice that at solar minimum over an equivalent one-year window.

Together, these results provide a more complete, physically grounded basis for CubeSat deorbit mission planning than either the analytical or numerical approach alone, and identify specific, stated simplifications — most notably the exclusion of the J2 perturbation — as natural directions for further extension. This work was developed in direct response to technical feedback from Dr. Dario Modenini (University of Bologna), whose three suggested extensions are each addressed in Sections 4.2.1 – 4.2.3.

6. References

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